Active Directory & Splunk – SOC Detection Lab

1. Project Overview

This project involved building a small Windows Active Directory environment and integrating it with Splunk Cloud to monitor and investigate Windows security events.

The objective was to understand how Security Operations Centre (SOC) analysts use centralized log monitoring and detection searches to identify potentially suspicious activities within a Windows domain environment.

The lab simulated common enterprise security events, including failed logins, successful logons, account creation, account modifications and privileged group membership changes.

2. Objectives

- Configure a Windows Server environment as an Active Directory Domain Controller.
- Create a domain and test user account.
- Install and configure Splunk Universal Forwarder.
- Forward Windows Event Logs to Splunk Cloud.
- Develop SPL searches to investigate security events.
- Create and test an alert for repeated failed login attempts.
- Understand the significance of account and group membership changes.

3. Lab Environment and Tools

Component	Configuration
Host operating system	Windows 11 Home
Virtualization	Oracle VirtualBox 7.2.8
Server environment	Windows Server VM
Domain	CYBERLAB
Domain Controller	AD-DC01
Test account	testuser
Log forwarding	Splunk Universal Forwarder
SIEM platform	Splunk Cloud
Log sources	Windows Security, System and Application

4. Lab Architecture

The lab consisted of a virtualized Windows Server environment configured as an Active Directory Domain Controller. Splunk Universal Forwarder was installed on the server to collect Windows Event Logs and transmit them to Splunk Cloud.

Log collection flow:

Windows Server / Active Directory → Splunk Universal Forwarder → Splunk Cloud → SPL Searches → Detection and Alerting

5. Active Directory Configuration

Active Directory Domain Services was configured on the Windows Server virtual machine.

The domain `CYBERLAB` was established, with the server operating as `AD-DC01`.

A test organizational unit named `Lab_Users` was created, followed by a test user account named `testuser`.

This environment was used to generate and investigate account-related security events.

6. Splunk Universal Forwarder Configuration

Splunk Universal Forwarder was installed on the Windows Server VM and configured to forward Windows Event Logs to Splunk Cloud.

The following inputs were enabled:

- Windows Security Event Log
- Windows System Event Log
- Windows Application Event Log

The configuration was verified using Splunk's `btool inputs list --debug` command.

Successful event ingestion was confirmed through searches in Splunk Cloud.

7. Security Monitoring and Detection Use Cases

7.1 Failed Login Attempts – Event ID 4625

Event ID 4625 records failed logon attempts.

A detection search was developed to identify repeated failed logins within a five-minute time window.

SPL search:

```
index=main EventCode=4625
| bin _time span=5m
| stats count by _time, Account_Name
| where count >= 5
```

Five incorrect password attempts were generated during testing. Splunk successfully received the corresponding failed-login events for `testuser`.

An alert titled **Brute Force – Multiple Failed Logins** was configured to run every minute, using a five-minute search window.

The alert successfully triggered during the validation test.

Finding: The detection successfully identified the threshold of repeated failed login attempts. This demonstrates a basic capability for identifying potential brute-force activity. Repeated failures alone do not establish that an actual attack occurred.

7.2 Successful Logon Monitoring – Event ID 4624

Event ID 4624 records successful logon activity.

The search returned approximately 3,230 events during the investigation.

The following logon types were observed:

Logon Type	Description
0	System
2	Interactive
3	Network
5	Service
7	Workstation unlock

Finding: The investigation demonstrated the ability to monitor successful logon activity and distinguish between different Windows logon types. Usable username extraction was limited in the inspected records.

7.3 User Account Creation – Event ID 4720

Event ID 4720 records the creation of a user account.

Four events were returned in the investigation.

The test account `testuser` was identified as having been created by the `Administrator` account within the `CYBERLAB` domain.

Other observed account-creation events included `WDAGUtilityAccount` and `krbtgt`.

These accounts were treated with appropriate context because Windows and Active Directory can generate or maintain system-managed accounts.

Finding: Account creation monitoring successfully identified the creation of the test account and provided information about the account creator and target account.

7.4 User Account Modification – Event ID 4738

Event ID 4738 records changes to user accounts.

The initial investigation returned 21 events. Target-account extraction identified activity involving:

- WDAGUtilityAccount – 6 events
- Administrator – 4 events
- krbtgt – 4 events
- testuser – 3 events
- DefaultAccount – 2 events
- Guest – also appeared in the results

One inspected event showed `Administrator` as the subject and `testuser` as the target account. The changed-attributes section was not populated in that event.

Finding: Account modification activity was successfully detected. However, the specific attributes changed were not available in the inspected event.

7.5 Global Security Group Membership Changes – Event ID 4728

Event ID 4728 records the addition of a member to a security-enabled global group.

Four events were returned.

The investigation identified the Administrator account being added to:

- Group Policy Creator Owners
- Domain Admins

These groups have important administrative privileges within Active Directory.

The inspected records showed `ANONYMOUS LOGON` as the subject. This was documented as an observation requiring contextual interpretation rather than evidence of unauthorized activity.

Finding: The investigation demonstrated how group membership events can be used to monitor changes involving privileged Active Directory groups.

7.6 Local Security Group Membership Changes – Event ID 4732

Event ID 4732 records the addition of a member to a security-enabled local group.

Fourteen events were returned.

The investigation identified changes involving groups such as:

- Administrators
- Users
- Guests
- IIS_IUSRS
- Windows Authorization Access Group
- Pre-Windows 2000 Compatible Access
- Denied RODC Password Replication Group

One important event showed the Domain Admins group being added to the built-in local Administrators group.

This is consistent with the expected administrative configuration of a domain controller.

Finding: Group membership monitoring provided visibility into changes to local security groups and privileged access configurations.

7.7 Account Disabled – Event ID 4725

The search returned no events.

Finding: No account-disable activity was observed in the available dataset during the investigation.

7.8 Account Lockout – Event ID 4740

The search returned no events.

Finding: No account-lockout activity was observed in the available dataset during the investigation.

7.9 Account Deletion – Event ID 4726

The search returned no events.

Finding: No account-deletion activity was observed in the available dataset during the investigation.

8. Alert Configuration and Validation

A scheduled alert was configured for repeated failed login attempts.

Setting	Configuration
Alert name	Brute Force – Multiple Failed Logins
Schedule	Every minute
Cron expression	`*/1 * * * *`
Search time range	Previous 5 minutes
Detection threshold	5 or more failed attempts
Expiration	365 days
Validation	Successfully triggered

The alert was tested by generating repeated incorrect password attempts for the test account.

The corresponding Event ID 4625 events appeared in Splunk, and the alert was subsequently observed in Triggered Alerts.

9. Key Findings

The project successfully demonstrated centralized Windows event collection and security monitoring through Splunk Cloud.

The main findings included:

1. Successful collection of Windows Security, System and Application logs.
2. Identification of repeated failed login attempts through Event ID 4625.
3. Successful validation of a scheduled failed-login alert.
4. Monitoring of successful logon activity through Event ID 4624.
5. Identification of account creation and modification events.
6. Visibility into global and local security group membership changes.
7. Identification of privileged group membership activity involving Domain Admins.
8. Recognition that system-generated events and expected administrative configurations must be distinguished from suspicious activity.

10. Limitations

- The lab used a small virtualized environment rather than a production enterprise network.
- Some Windows event fields, particularly account names and changed attributes, were not consistently extracted or populated.
- Certain group membership events contained unresolved account or group names.
- The absence of an event in the available dataset does not establish that the corresponding activity never occurred.
- The brute-force detection was validated using manually generated failed-login attempts; no real brute-force attack was conducted.
- The alert represents a basic threshold-based detection and would require additional correlation and tuning for production use.

11. Conclusion

This project provided practical experience in configuring an Active Directory environment, forwarding Windows Event Logs to Splunk Cloud and developing basic security detection searches.

Through the investigation of Windows Security Event IDs, the lab demonstrated how centralized log monitoring can help identify authentication activity, account changes and privileged group membership modifications.

The successful validation of the failed-login alert demonstrated the practical use of SPL-based detection and scheduled alerting.

Overall, the project strengthened practical understanding of Windows security events, SIEM monitoring, log analysis and foundational SOC investigation techniques.

12. Future Improvements

- Develop additional correlation-based detection rules.
- Improve field extraction and event normalization.
- Monitor PowerShell activity and suspicious process execution.
- Investigate privileged account usage and abnormal authentication patterns.
- Build Splunk dashboards for authentication and account activity.
- Extend the lab with additional domain-joined machines and attack simulations.